WEEK 6 / LESSON 2 / 50 MINUTES

L2

Docker & Kubernetes

Docker ended the joke in 2013. K8s ended the "one-server-per-app" era in 2014. By 2026, no Indian AI deployment worth the name runs outside a container orchestrator.

Which of your models runs on a container today? Which doesn't?

WHAT'S INSIDE

WHY THIS LESSON MATTERS

"Works on my machine" is a punchline. Make it a joke.

WHY
Real
A real reason this matters in industry today.

SCALE
Today
A real reason this matters in industry today.

SPEED
Now
A real reason this matters in industry today.

IMPACT
Yours
A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1. Production Dockerfile for Python ML service.
2. Build, tag, push to registry.
3. Pod · Deployment · Service.
4. Rolling updates + rollbacks in K8s.
5. Deploy a model to managed K8s.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01
Production Dockerfile — 7...
Anatomy of a shipped image

02
K8s primitives — three you...
Key concept of this lesson.

03
Readiness probes — mandatory
Without them, rolling updates serve half-booted pods

04
Rolling updates + rollbacks
Zero-downtime deploys

01

PART 1 OF 4

Production Dockerfile — 7...

Anatomy of a shipped image

CONCEPT 1 OF 4

Production Dockerfile — 7 rules

1. Slim base image.
2. Non-root user.
3. Cache deps layer (requirements first).
4. Pin ALL versions.
5. HEALTHCHECK.
6. Single responsibility.
7. No secrets in image.

RULES
7 checks

Print on the wall of the infra pod.

DEFINITION

Production Dockerfile — 7 rules

Anatomy of a shipped image

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

K8s primitives — three you need

A core idea you'll use throughout this lesson.

KEY POINTS

- A core idea you'll use throughout this lesson
- Practical, named, applied to a real Indian use-case.
- Practical, named, applied to a real Indian use-case.

CONCEPT 3 OF 4

Readiness probes — mandatory

Kubelet polls /health; only routes traffic when pass.
Liveness probe restarts pods that hang.
Startup probe for slow-warming apps.
Every deployment needs all three for production.
NON-NEGOTIABLE
3 probes

readiness + liveness + startup.

DEFINITION

Readiness probes — mandatory

Without them, rolling updates serve half-booted pods

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Rolling updates + rollbacks

kubectl set image → one pod at a time.
Readiness probes gate new pods.
kubectl rollout undo → previous stable.
kubectl rollout history → the audit trail.
ZERO DOWNTIME
By default

Only if probes are correct.

KEY POINTS

- kubectl set image → one pod at a time
- Readiness probes gate new pods
- kubectl rollout undo → previous stable
- kubectl rollout history → the audit trail

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK · PAIR · SHARE

ACTIVITY

Build + run in class

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1.

2.

3.

4. Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1	*Which of the 7 Dockerfile rules do you break most?*
Q2	*Which probe is missing from your current deploy?*
Q3	*If your company goes all-in on one cloud, which K8s service?*

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

KNOWLEDGE CHECK · QUESTION 1 OF 3

Best practice Dockerfile:

- **A** Root user
- **B** Non-root + pinned + healthcheck
- **C** Secrets baked in
- **D** No healthcheck

ANSWER · QUESTION 1 OF 3

Best practice Dockerfile:

CORRECT ANSWER

B

Non-root + pinned + healthcheck

WHY

Defence in depth + reproducibility + observability

KNOWLEDGE CHECK · QUESTION 2 OF 3

Stable network endpoint for pods:

- **A** Pod
- **B** Deployment
- **C** Service
- **D** Namespace

ANSWER · QUESTION 2 OF 3

Stable network endpoint for pods:

C

CORRECT ANSWER

Service

WHY

Service gives a stable DNS / IP across pod churn

KNOWLEDGE CHECK · QUESTION 3 OF 3

Why readiness probes:

- **A** Style
- **B** Gate traffic to ready pods
- **C** Scale
- **D** UI

ANSWER · QUESTION 3 OF 3

Why readiness probes:

CORRECT ANSWER

B

Gate traffic to ready pods

WHY

Otherwise rolling updates serve half-booted pods

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

1. Slim base image.
2. Non-root user.
3. Cache deps layer (requirements first).
4. Pin ALL versions.
5. HEALTHCHECK.
6. Single responsibility.
7. No secrets in image.

RULES
7 checks
Print on the wall of the infra pod.

THE TAKEAWAY

"Docker + K8s + probes. The operational spine of 2026 AI."

WHAT TO NOTICE

1 Production Dockerfile — 7...

2 K8s primitives — three you...

3 Readiness probes — mandatory

4 Rolling updates + rollbacks

APPLY IT → ASSIGNMENT

Containerise + deploy locally

Your W3 prediction API, containerised with the 7 rules + deployed to kind.

DELIVERABLES

- Dockerfile with 7 rules.
- Deployment YAML with 3 probes.
- Service of type LoadBalancer.
- Rolling update from v1 to v2.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Repo URL + 30-s screen recording.

ONE THING TO REMEMBER

> *"Docker + K8s + probes. The operational spine of 2026 AI."*

— AI Learning Hub

END OF LESSON 6

What you can now do

Each one of these is now part of your working vocabulary.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

NEXT → Lesson 3 — Cloud platf orms.